

Problem 1

- Let $A = \bigcap_{\alpha \in J} \mathcal{T}_\alpha$
- Since for all $\mathcal{T}_\alpha$ s.t. $\alpha \in J$ are topologies they all have $X \in \mathcal{T}_\alpha$ and $\emptyset \in \mathcal{T}_\alpha$ thus $X, \emptyset \in A$
- Let $U, V \in A$. Thus, $U, V \in \mathcal{T}_\alpha$ for all $\alpha \in J$ thus since these are topologies this implies that $U \cap V \in \mathcal{T}_\alpha$ for all $\alpha \in J$ thus $U \cap V \in A$
- Let $U = \bigcup_{i \in I} U_i$ s.t. $U_i \in A$ and I is an indexing set. Since each $U_j \in A$ this means it is in every $\mathcal{T}_\alpha$. Since these are topologies they are closed under arbitrary unions thus $U \in \mathcal{T}_\alpha$ for all $\alpha \in J$ thus $U \in A$.

Thus, A is a topology on X .

Counter Example:

Let $X = \{0, 1, 2\}$. Define two topologies on X :

- $\mathcal{T}_1 = \{\emptyset, X, \{0\}, \{1, 2\}\}$
- $\mathcal{T}_2 = \{\emptyset, X, \{1\}, \{0, 2\}\}$

It is straightforward to check each is a topology.

Now consider

$$\mathcal{T}_1 \cup \mathcal{T}_2 = \{\emptyset, X, \{0\}, \{1, 2\}, \{1\}, \{0, 2\}\}$$

However:

$$\{1, 2\}, \{0, 2\} \in \mathcal{T}_1 \cup \mathcal{T}_2 \quad \text{but} \quad \{1, 2\} \cap \{0, 2\} = \{2\} \notin \mathcal{T}_1 \cap \mathcal{T}_2$$

So it is not true that unions of topologies are always topologies.

Problem 2

Assume $U_1, \dots, U_n$ is a finite collection of open subsets of a topological space X .

Base Case:

Let $n = 2$, and let U_1, U_2 be open. Thus, $U_1 \cap U_2$ are open in X by the definition of a topology.

Inductive Step:

Assume for all $n \leq k$ s.t. $2 \leq k$ and $n, k \in \mathbb{N}$ the union of n open sets is open.

Consider $k + 1$:

Let $U_1, \dots, U_{k+1}$ be open. Now consider $U_1 \cap \dots \cap U_k$ by the inductive hypothesis this is open call this set U . Thus, consider $U \cap U_{k+1}$ by the definition of topology the intersection of two open sets is open. Thus, by induction the intersection of n open sets of a topological space is open for all $n \in \mathbb{N}$

Problem 3

- Let $\{C_\alpha\}_{\alpha \in J}$ be a collection of closed sets in X with indexing set J . Thus, by definition of closed the complement is open thus $C_\alpha = X \setminus O_\alpha$ for some open set O_α . Consider:

$$\bigcap_{\alpha \in J} C_\alpha = \bigcap_{\alpha \in J} (X \setminus O_\alpha)$$

By De Morgan's Law:

$$= X \setminus \bigcup_{\alpha \in J} O_\alpha$$

Thus since open sets are closed under arbitrary union that is an open set thus since the complement of that intersection is open we know that the intersection is closed.

- – Base Case: Let $n = 2$. Let C, V be closed. Thus $C = X \setminus O$ and $V = X \setminus U$ for open sets O, U . Consider:

$$C \cup V = (X \setminus O) \cup (X \setminus U)$$

By De Morgan's Law:

$$= X \setminus (O \cap U)$$

Since $O \cap U$ is open we know that the complement of $C \cup V$ is open this $C \cup V$ is closed.

- Inductive Step: Assume for all unions of closed sets of less than or equal to $k \geq 2$ is closed
Consider $k + 1$ sets. Let $C_1 \dots C_{k+1}$ be closed. Thus consider $C_1 \cup \dots \cup C_k$. By the inductive hypothesis this is closed. Thus call this set C . Thus, consider $C \cap C_{k+1}$ by the base case it is also closed. Thus, unions of closed sets of size n is closed for all $n \in \mathbb{N}$.

Let $X = \mathbb{R}$ and use the standard topology on $\mathbb{R}$. Consider the collection $U_n = [\frac{1}{n}, 1]$ for $n \in \mathbb{N}$. These are by definition closed since it is the complement of $(-\infty, \frac{1}{n}) \cup (1, \infty)$.

Consider:

$$U = \bigcup_{n \in \mathbb{N}} U_n$$

Let $x \in (0, 1]$ thus by the archimedian property there is $n \in \mathbb{N}$ s.t. $1/n < x$ thus $x \in U_n \subseteq U$
Let $x \in U$ thus $x \in U_n$ for some $n \in \mathbb{N}$ thus since $[1/n, 1] \subset (0, 1]$ thus $x \in (0, 1]$ thus $U = (0, 1]$

However since $\mathbb{R} \setminus U = (-\infty, 0] \cup (1, \infty)$ is not open. This means that U is not closed. Thus, this is an example of how closed sets are not closed under arbitrary union.

Problem 4

Consider $H = \{(x, y, z) \in \mathbb{R}^3 \mid z > 0\}$. Let $p \in H$. Let $\epsilon = z/2 > 0$

Consider the ball $B_\epsilon(p)$

Let $q \in B_\epsilon(p)$ thus $q = (x', y', z')$. To show that this is in H all we need to show is that $z' > 0$.

From the definition of a ball we know that $|z' - z| < \epsilon \implies z - \epsilon < z' \implies z - z/2 = z/2 < z'$. Thus since $z > 0$ we know that $z/2 > 0$ thus $0 < z'$. Thus $q \in H$.

Thus, for every point we can find a ball that is a subset of H . Thus, H is an open subset of $\mathbb{R}^3$

Problem 5

- (a)
- Let $x \in X$ since X is the unions of sets in $\mathcal{S}$ there exists an $x \in S$ s.t. $S \in \mathcal{S}$. Thus, since $S = S \cap S$, x is in the union of these finite intersections.
 - Let U_1, U_2 be in the set of the union of these finite intersections. Thus, $U_1 = S_1 \cap S_2 \cap \dots \cap S_n$, and $U_2 = T_1 \cap T_2 \cap \dots \cap T_m$ for some $S_i, T_j \in \mathcal{S}$ for $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, m\}$

Consider $U_1 \cap U_2$:

Let $x \in U_1 \cap U_2$. Consider $U_1 \cap U_2 = U_3$. Since U_1 and U_2 are finite intersections this means U_3 is a finite intersection thus it is in our set. Thus $x \in U_3 \subseteq U_1 \cap U_2$

Thus this set is a basis for a topology on X .

- (b) Let $\{\mathcal{T}_\alpha\}_{\alpha \in J}$ be all the topologies that contain $\mathcal{S}$. Let

$$\mathcal{T} = \bigcap_{\alpha \in J} \mathcal{T}_\alpha$$

Let B be the topology generated by the basis in (a).

$\mathcal{T}$ is the intersection of every topology that contains $\mathcal{S}$ and B contains $\mathcal{S}$ we know that $\mathcal{T} \subseteq B$.

Since B is the collection of all unions of finite intersections of elements of $\mathcal{S}$, and from problem 1 we know it is a topology. This means we are closed under arbitrary unions and finite intersections thus every element of B is in $\mathcal{T}$.

Thus, $\mathcal{T} = B$

Thus we know that B is the coarsest topology that contains $\mathcal{S}$.

Problem 6

- (a) Let β_d be the basis for the order topology on $\mathbb{R} \times \mathbb{R}$ with dictionary order, and let β_p be the basis for the product topology.

Let $B \in \beta_p$. WLOG let $B = (a, b) \times (c, d)$ since we can create any basis element from unions or intersections of this type of element. Let $p \in B$ s.t. $p = (x, y)$. Consider the basis element $C = ((x, y - \frac{y-c}{2}), (x, y + \frac{d-y}{2}))$. Thus, $p \in C \subseteq B$, and C is clearly a basis element for the order topology. Thus, we have shown that the order topology is finer than the product topology.

Consider the set $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x = 0 \text{ and } 0 < y < 1\}$. This is open in the order topology since it corresponds to the interval $(0, 0) < (0, y) < (0, 1)$. However, this is not open in the product topology, since it is equivalent to the set $A = \{0\} \times (0, 1)$, and $\{0\}$ is not open in $\mathbb{R}$. Thus, A is not open in the product topology.

- (b) Consider the point $p = (x, y) \in \mathbb{Z} \times \mathbb{Z}$. Consider the open set in the order topology: $((x, y - 1), (x, y + 1)) = \{(x, y)\}$. Thus, every singleton is actually an open set in $\mathbb{Z} \times \mathbb{Z}$. Thus, we can write every subset as a union of singletons thus we are actually in the discrete topology since every subset is open. Consider the open set in the product topology $(x - 1, x + 1) \times (y - 1, y + 1) = \{p\}$. Again we have shown all singletons are open thus all subsets are open thus we are in the discrete topology. Thus, these two topologies are the same.

- (c) Consider the point $p = (x, y) \in \mathbb{N} \times \mathbb{N}$. If $x = y = 0$ then consider the open set $[0, 1) \times [0, 1) = \{p\}$. Thus, this singleton is open. Now if $x = 0$ consider the open set $(x - 1, x + 1) \times [0, 1) = \{p\}$ now we know this singleton is open and do the same for if $x \neq 0$ and $y = 0$. If they are both not zero then consider the open set $(x - 1, x + 1) \times (y - 1, y + 1) = \{p\}$. Thus, all singletons are open thus by the same argument as above we are in the discrete topology. Since the discrete topology is equal to all the subsets of $\mathbb{N} \times \mathbb{N}$. This means that any other topology will be coarser than it since any collection of subsets is a subset to the power set. Thus, the dictionary order topology is coarser than the product topology.

Consider the open set $\{(1, 0)\}$ in the product topology. For this to be open in the order topology we need a basis element s.t. $((a, b), (c, d)) = \{(1, 0)\}$. If $a = 0$ then we will have all elements $(0, b + 1), (0, b + 2), \dots$. Thus, $a \neq 0$. If $c > 1$, then we will have all elements of the form (c, x) s.t. $x < d$. Thus, $c > 1$. Thus, $a = 1 = c$, since $a \leq 1 \leq c$. Thus, now consider b, d . If $d > 1$ then we would have the element $(1, 1)$ thus $d = 1$. Thus we have the interval $((1, b), (1, 1))$. For this set to only have $(1, 0)$ this would mean that $d < 0$. However that cannot happen since $d \in \mathbb{N}$. Thus this singleton is not open in the order topology. Thus, these two topologies are not equal.

Problem 7

- (a) Let X be a finite metrizable space. Let d be a metric on X that induces a topology on X . Let $X = \{x_1, \dots, x_n\}$. Thus define r_i for each $i \in \{1, \dots, n\}$ to be the minimum value of $\{d(x_i, x_j) \mid i \neq j \in \{1, \dots, n\}\}$. There exists a min since we only have a finite amount of points thus we will only have $n - 1$ values. By the definition of metrizable we know that $r_i > 0$ for all i . Thus consider the open set $B(x_i, r_i/2) = \{x \in X \mid d(x_i, x) < r_i/2\}$. Since $r_i/2 < r_i \leq d(x_i, x_j)$ for $j \neq i$. This ball only contains the point x_i . Thus we have shown that all singletons are open sets. Thus we can unions any pair of singletons to get any subset of X . Thus, we know that all open sets are open. Thus, it is discrete.
- (b) Look at $\mathbb{Q}$ with standard topology. Consider the set $\{1\}$ By definition of the standard topology on $\mathbb{Q}$ for $\{1\}$ to be open there must exists a $\epsilon > 0$ s.t. $B(1, \epsilon) = \{x \in \mathbb{Q} \mid |x - 1| < \epsilon\} = (1 - \epsilon, 1 + \epsilon) = \{1\}$. However, by the archimedian property there exists a $n \in \mathbb{N}$ s.t. $1/n < \epsilon$. Thus consider the value $1 + 1/n \in \mathbb{Q}$. Thus since $1 - \epsilon < 1 + 1/n < 1 + \epsilon$ this value is in our ball thus there is no $\epsilon > 0$ s.t. 1 is the only thing in the set. Thus, $\{1\}$ is not open. Thus, $\mathbb{Q}$ is not discrete under the standard topology.
- (c) Let X be a discrete topological space. Consider:

$$d(x, y) = \begin{cases} 0, & \text{if } x = y, \\ 1, & \text{if } x \neq y. \end{cases}$$

- $d(x, y) = 0 = d(y, x) \iff x = y$ this is given from the definition of d .
- Assume $y \neq x$ and $x, y \in X$. Thus $d(x, y) = 1 = d(y, x)$
- Let $x, y, z \in X$.

Assume $x \neq z$ since if this is true then $d(x, z) = 0 \leq d(x, y) + d(y, z)$ is always true since d is always 0 or 1.

If $x = y \neq z$ then $d(x, z) = 1 \leq d(x, y) + d(y, z) = 1$

If $x \neq y \neq z$ then $d(x, z) = 1 \leq d(x, y) + d(y, z) = 2$

Thus d is a metric.

Thus consider the basis for the metric topology. Let $x \in X$, then consider the ball $B_\epsilon(x)$ for $\epsilon = 1/2$. Since all distances are 1 there will only be one thing in this ball and that is x thus we know that all singletons are open. Thus, we can get any subset by unions of singletons. Thus, all subsets of X are open. Thus X is metrizable since we can induce the same topology with a metric.